Governance of AI-Based Algorithms

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Abstract

As artificial intelligence (AI)-based algorithms continue to transform various sectors, from healthcare and finance to social media and governance, the need for robust oversight mechanisms has become increasingly critical. This chapter explores the governance of AI-based algorithms, defining it as “the processes, policies, and frameworks established to regulate and oversee the development, deployment, and impact of AI-based algorithms to ensure they are ethical, transparent, and aligned with societal values.” The chapter emphasizes a human-centered AI perspective, advocating for ethical, transparent, fair, and accountable AI governance that prioritizes societal well-being. The discussion begins by providing an overview of AI-based algorithms, their applications, and the challenges they present, including bias, opacity, and accountability gaps. It then examines the fundamental principles necessary for governing AI-based algorithms, including transparency, accountability, fairness, privacy, and safety. Additionally, the chapter highlights regulatory and oversight frameworks designed to mitigate AI-related risks while fostering responsible innovation. By integrating human-centered governance approaches, this chapter underscores the importance of aligning AI algorithms with human values and societal needs. The analysis also examines mechanisms for evaluating AI models, ensuring ethical deployment, and establishing accountability structures that safeguard against potential harm. Ultimately, the chapter contributes to the broader discourse on AI governance, offering insights into how responsible and equitable AI systems can be developed and implemented in alignment with global ethical standards.

Keywords

ai governance · Privacy · AI safety · Transparency · Accountability · Algorithmic bias

1 Introduction

Artificial intelligence (AI) algorithms have become widely integrated for use within various domains, shaping decision-making processes and highlighting an imperative for pragmatic regulatory oversight. AI-based algorithms refer to computational models designed to analyze data, identify patterns, and generate predictions or decisions autonomously. These algorithms power applications ranging from healthcare diagnostics and autonomous vehicles to financial forecasting and recommendation systems. In healthcare, AI-based algorithms assist in disease detection

(Bejnordi et al., 2017; Yoo et al., 2020), personalized treatment plans (Carini & Seyhan, 2024), and robotic-assisted surgeries (Knudsen et al., 2024). In finance, AI-enabled algorithms can be leveraged for fraud detection (Bao et al., 2022), algorithmic trading (Beverungen, 2019), and credit risk assessments (Shi et al., 2022). Social media platforms leverage AI-driven algorithms to personalize content recommendations, moderate harmful posts, and analyze user sentiment (Kim, 2017). Despite the advantages of leveraging AI for these use cases, AI-based algorithms also present risks, such as bias, lack of transparency, and ethical dilemmas (Bengio et al., 2025), necessitating structured governance mechanisms.

The governance of AI-based algorithms refers to the processes, policies, and frameworks established to regulate and oversee the development, deployment, and impact of AI-based algorithms to ensure they are ethical, transparent, and aligned with societal values (Nitzberg & Zysman, 2022). Effective governance mechanisms establish accountability, ensure compliance with ethical standards, and mitigate potential harms associated with AI implementation. A human-centered approach to AI governance prioritizes the ethical and responsible development of AI-based algorithms, ensuring they align with fundamental human values, fairness, and transparency (Sigfrids et al., 2023). Human-centered AI governance emphasizes the need for AI systems to be interpretable, inclusive, and subject to oversight mechanisms that hold developers and organizations accountable for their impacts. This perspective advocates for stakeholder engagement, interdisciplinary collaboration, and continuous monitoring to align AI systems with societal expectations and public interest.

Governance is essential for AI-based algorithms due to the risks and challenges they present, such as bias, lack of transparency, and accountability gaps. Without proper oversight, AI systems can reinforce societal inequalities and produce harmful outcomes. Thus, developers, researchers, and organizations must be dedicated to mitigating AI concerns, along with supporting efforts by governments to draft and enact AI legislation. To move toward effective governance frameworks, stakeholders across the AI research and development (R&D) lifecycle must work to make AI-based algorithms transparent and interpretable to foster trust and facilitate accountability. To promote algorithmic fairness, governance mechanisms must address bias in AI training data and decision-making processes to ensure equitable outcomes. Additionally, developers, organizations, and regulators should establish clear accountability structures to mitigate potential harms from AI applications. Overall, algorithmic systems must be aligned with human values and needs, and comprehensive AI governance can ensure that algorithmic decisions reflect ethical considerations and respect human rights, prioritizing social good over profit-driven objectives.

This chapter explores the fundamental principles in governing AI-based algorithms, detailing critical aspects such as transparency, accountability, fairness, privacy, and safety. Procedural rights in AI governance encompass mechanisms that ensure due process in algorithmic decision-making, with explainability requiring that system operations be intelligible to affected parties, while accountability establishes clear responsibility pathways when systems cause harm. These procedural protections fundamentally enable the meaningful exercise of substantive rights like privacy and nondiscrimination by providing the necessary mechanisms through which individuals can recognize, challenge, and remedy algorithmic harms. There is a crucial distinction in the relationship between regulation and governance, where regulation typically constitutes formal, legally binding rules enforced by designated authorities, while governance encompasses the broader ecosystem of norms, standards, incentives, and practices that shape algorithmic development and deployment, including industry self-regulation, technical standards, and institutional policies that often precede or complement formal regulation. This chapter examines various regulatory and oversight frameworks designed to mitigate the risks associated with AI-based algorithms while fostering responsible innovation. The chapter also discusses human-centered approaches to AI governance and outlines mechanisms for evaluating AI models to ensure ethical deployment and use. By providing a comprehensive analysis, the chapter aims to contribute to the ongoing discourse on governing AI-enabled algorithms and the necessity of aligning AI technologies with societal values.

2 Fundamental Principles in Governing AI-Based Algorithms

To lay the groundwork for effective oversight, it is essential to establish mechanisms that can guide the development and deployment of AI-based algorithms. Outlining fundamental principles for governing AI-based algorithms is essential because these sociotechnical systems increasingly mediate critical domains of human decision-making and social organization while operating through complex, often opaque mechanisms that can perpetuate or amplify societal biases, evade traditional accountability structures, and generate novel ethical challenges that existing regulatory frameworks are ill-equipped to address. This section details some of the core principles necessary for governing AI-based algorithms such as transparency, explainability, bias, privacy, accountability, and reliability.

2.1 Transparency and Explainability

Over the past decade, a growing amount of research and policy documents have focused on improving transparency in AI systems to understand how these systems work, what data is used to train them, and how they make decisions. Transparency is fundamental to governing AI algorithms as it enables stakeholders to understand how decisions are made and provides a foundation for accountability (Coglianese & Lehr, 2019; Larsson & Heintz, 2020). When the mechanisms of AI systems are opaque, trust erodes, particularly in high-stakes environments where decisions significantly impact individuals and communities. Transparency involves clear documentation of data sources, model architecture, training processes, and decision-making logic. This documentation allows regulators, researchers, and the public to scrutinize AI systems effectively. AI technologies continue to be rapidly integrated into many facets of daily life, including public services, social media, financial management, and medical care.

AI explainability (also referred to as XAI) complements transparency by translating complex algorithmic processes into understandable terms (Dwivedi et al., 2023). Explainability ensures that stakeholders, including end users, can comprehend why a particular decision or prediction was made. This is especially critical in domains like healthcare, where understanding the rationale behind a diagnosis or treatment recommendation can improve both trust and outcomes (Gerlings et al., 2022). While explainability has often been touted as a way to increase transparency, a significant amount of XAI methods are uninterpretable to stakeholders without technical domain expertise and often add additional layers of complexity to ML models (de Bruijn et al., 2022; Bao and Zeng, 2024). These issues necessitate a shift toward the development of more interpretable XAI methods that are accessible to stakeholders with varying amounts of AI literacy and technical knowledge, along with ensuring such approaches are domain-relevant (Okolo, 2023). By fostering trust through explainability, organizations can also enhance accountability, ensuring that decision-makers are held responsible for the impacts of AI systems.

2.2 Bias, Fairness, and Underrepresentation

Bias in AI algorithms poses a significant challenge to fairness and equity, particularly in domains like hiring, criminal justice, and lending (Kordzadeh & Ghasemaghaei, 2022; Bengio et al., 2025). This can lead to discrimination in hiring (Andrews & Bucher, 2022; Kassir et al., 2023), negative healthcare outcomes (Obermeyer et al., 2019), lower access to financial resources (Kelley et al., 2022; Garcia et al., 2024), and disparate legal outcomes (Angwin et al., 2022). Algorithmic bias often arises from imbalances or prejudices in training data, which can perpetuate or amplify existing societal inequities. Bias can often exist in the way machine learning algorithms are designed and also how they are integrated into the architectures of AI systems, furthering the impacts of skewed data. Methods such as Reinforcement Learning from Human Feedback (RLHF), which aims to align model decision-making with human preference, have been introduced to mitigate bias (Ouyang et al., 2022). However, these approaches come with limitations, which could lead to the inadvertent introduction of new biases (Santurkar et al., 2023; Hartmann et al., 2023) and enable AI models to incorporate less factual information (Sharma et al., 2024). To address bias, governance frameworks must prioritize diverse and representative data collection practices. Additionally, rigorous testing and auditing processes are essential to identify and mitigate bias throughout the AI lifecycle (Raji et al., 2020).

Fairness enhances motivations to eliminate bias and prioritize equitable access and outcomes for all demographic groups. Datasets used to train and evaluate AI models are predominantly collected in English and represent Western cultures, with a bias toward the United States popular culture (Longpre et al., 2023). These biases are likely to continue given that 52.1% of all web content on the Internet is estimated to be in English, with Spanish, German, Russian, and Japanese taking up 5.5%, 4.8%, 4.5%, and 4.3%, respectively (Statista, 2024). These issues of underrepresentation are further exacerbated by the types of subject matter included in online content, which are also less likely to represent communities whose cultures and histories have not been captured in written formats. Additionally, transcribing oral archives into text can lose subjective contextual cues (Pessanha & Salah 2021), which can also lead to potential biases in datasets. Moving toward more inclusive representation will require adopting inclusive design principles for dataset collection (Scheuerman et al., 2021) and AI development, prioritizing stakeholder engagement, and incorporating participatory methods during development. Additionally, developers and AI companies must address the underrepresentation of marginalized groups in training datasets to prevent the systematic exclusion of vulnerable populations. To assist in this process, governance frameworks should mandate that organizations assess the social impact of their algorithms and incorporate fairness metrics into evaluation criteria.

2.3 Privacy

Privacy is a fundamental human right that must be protected in the development and deployment of AI algorithms. The rise of tools like ChatGPT and the heightened interest in developing AI have led to increasing reliance on vast amounts of personal data to train ML algorithms and operate AI systems (Sadowski, 2019). This has subsequently amplified privacy concerns and paved the way for increased data extraction from social media platforms, mobile applications, and websites. Given the patchwork set of state-level AI regulation and lack of federal data privacy regulation in the United States, there are fewer incentives for companies operating in this region to set strict data privacy standards for non-health and financial data, which are covered by the Health Insurance Portability and Accountability Act (HIPAA) and the 1978 Right to Financial Privacy Act (RFPA) and the Financial Modernization Act of 1999. The International Association of Privacy Professionals estimates that 144 countries have national data privacy laws (Apacible-Bernardo & Bushey, 2025). Additionally, governance frameworks by continental bodies, such as the European Union's General Data Protection Regulation (GDPR), play a critical role in regulating data collection, storage, and usage. The African Union has also introduced the Malabo Convention, which emphasizes the importance of transparency, accountability, and human oversight in AI systems while also promoting the use of AI for the social and economic advancement of African nations.

Under privacy laws like GDPR, organizations are required to obtain explicit consent for data usage, ensure data minimization, and provide individuals with the right to access and delete their data (European Parliament, 2016). To further privacy goals, companies and organizations should also be encouraged to adopt privacy-enhancing technologies, such as differential privacy and federated learning, to protect sensitive information while enabling AI innovation (Timan & Mann, 2021; Yanamala et al., 2024). By enforcing stringent privacy standards, governance frameworks can safeguard individuals from potential harm, such as unauthorized data sharing and surveillance. Comprehensively, privacy serves as a cornerstone principle in the governance of AI algorithms, operating at the intersection of individual rights, collective interests, and technical capabilities. The fundamental role of privacy in AI governance manifests through multiple dimensions: as a human right that must be protected, as a technical constraint that shapes algorithmic design, and as a social value that influences public trust in automated systems. This multifaceted conceptualization of privacy necessitates governance frameworks that address not only the direct collection and processing of personal data but also the emergent privacy implications of algorithmic inference capabilities. The sophisticated nature of modern AI-based algorithms, which are capable of extracting latent patterns and drawing unexpected correlations, has transformed privacy from a binary concept of data protection into a complex sociotechnical challenge that requires continuous negotiation between competing interests of innovation, utility, and individual autonomy (Susser et al., 2019).

2.4 Accountability and Liability

AI accountability encompasses a multifaceted approach that addresses transparency, fairness, bias mitigation, explainability, and regulatory compliance (Raja & Zhou, 2023). By embracing these principles and practices, organizations can ensure that AI-based algorithms are developed and deployed in a responsible and accountable manner, maximizing the benefits of AI while minimizing potential risks and harms. While there are ongoing debates surrounding the attribution of accountability for decisions generated by AI systems (Doshi-Velez et al., 2017; Novelli et al., 2024)—whether it lies with individual developers, the companies that deploy them, or the algorithms themselves—it remains absolutely crucial to establish clear governance structures and ethical guidelines for the development and deployment of AI-based algorithms. This includes implementing robust testing and validation procedures, ensuring transparency in AI decision-making processes, and establishing mechanisms for redress and accountability in cases where AI systems cause harm or produce unintended consequences. Regulatory compliance is another vital aspect of AI accountability, necessitating adherence to existing laws and regulations governing data privacy, security, and consumer protection (Oduro et al., 2022; Orio & Noor, 2024). Moreover, ongoing collaboration between policymakers, industry leaders, and ethicists is crucial to developing and adopting regulations that keep pace with the rapid advancements in AI technology.

AI liability, which builds upon established accountability measures, endeavors to allocate responsibility and repercussions for any harm or damage that may arise from the utilization of AI-based algorithms (Durante & Floridi, 2022). This encompasses a broad spectrum of potential impacts, including physical injuries, financial losses, and reputational damage. The attribution of liability in AI systems can be intricate due to the multifaceted nature of their development, deployment, and operation, which often involves multiple stakeholders. These stakeholders can include developers, manufacturers, service providers, and end users. To effectively navigate the complexities of AI liability, governments must develop comprehensive frameworks that are applicable to the respective domain of interest (healthcare, education, manufacturing, finance, etc.), delineate the roles and responsibilities of each stakeholder, establish clear standards for AI safety and performance, and provide mechanisms for redress in the event of AI-enabled harm. Furthermore, frameworks should be adaptable to the evolving nature of AI technology and its applications and must also address the complexity of assigning responsibility in cases where AI algorithms cause harm (Moch, 2024). For instance, if an autonomous vehicle is involved in an accident, such frameworks should be able to address whether the developer, manufacturer, or operator is liable. Governance frameworks should provide clarity on these issues by establishing standards for risk assessment, incident reporting, and redress mechanisms. Holding stakeholders accountable not only mitigates risks but also incentivizes the development of safer and more ethical AI systems.

2.5 Safety and Reliability

The safe and reliable operation of AI-based algorithms is critically important within broader AI governance, particularly in high-consequence domains such as healthcare, autonomous vehicles, and financial systems. Safety in AI-based algorithms surpasses conventional software safety paradigms, which primarily focus on implementing verification and validation processes (Suski et al., 1994). In the context of AI-based algorithms, safety encompasses both the immediate operational safety of algorithmic systems and their broader societal implications. At its core, AI safety demands robust mechanisms for preventing harmful outputs, ensuring algorithmic stability, and maintaining operational boundaries within predefined ethical and technical parameters (Díaz-Rodríguez et al., 2023; Chen et al., 2024). This includes implementing comprehensive safety protocols that address various failure modes, from technical malfunctions to adversarial attacks, while simultaneously considering the cascade effects of potential failures across interconnected systems. In critical applications, safety and reliability standards must align with industry-specific regulations in the healthcare, automotive, manufacturing, nuclear, and aviation industries (Lupo, 2023; Muhlheim et al., 2023). By enforcing these standards, governance frameworks for AI-based algorithms can minimize risks and build public confidence in the deployment of AI technologies. Incorporating AI safety into governance mechanisms requires a proactive approach that integrates risk assessment methodologies, establishes clear safety metrics, and implements continuous monitoring systems to detect and mitigate potential hazards within AI-based algorithms before they materialize.

Reliability in AI-based algorithms builds upon these foundational safety considerations by extending beyond mere prevention of harm to ensure consistent, dependable, and trustworthy algorithmic performance across diverse operational contexts (Ozkaya, 2020). While safety establishes the baseline for preventing adverse outcomes, reliability focuses on maintaining optimal performance and predictable behavior over time, even under varying conditions and inputs (Appelganc et al., 2022; Hong et al., 2023). This encompasses both technical reliability metrics, such as system uptime and accuracy rates, and broader socio-technical considerations, like the algorithm's ability to maintain its intended functionality across different social contexts and user populations (Ryan, 2020). Additionally, safety standards can play a role in helping AI-based algorithms align with industry-specific regulations, such as the ISO 26262 Standard for Automotive Safety (ISO, 2011) or the FDA's Draft Guidance for AI-based Device Software Functions (FDA, 2025). By enforcing these standards, governance frameworks for AI-based algorithms can minimize risks and build public confidence in the deployment of AI technologies. Governing reliability in AI-based algorithms must, therefore, address not only the technical aspects of system stability but also the social dimensions of algorithmic dependability (Avizienis et al., 2001), including transparency in performance metrics, clear protocols for system updates and maintenance, and established procedures for handling edge cases and exceptional situations. This comprehensive approach to reliability ensures that AI-based algorithms not only operate safely but also deliver consistent and trustworthy results that align with stakeholder expectations and societal values.

3 Governance Approaches for AI-Based Algorithms

Having explored the core principles that help set the foundation for governing AI-based algorithms, it is now crucial to examine how these principles translate into frameworks, guidelines, and standards, which have influenced various approaches to regulation globally. This section examines several key governance approaches designed to regulate AI-based algorithms, detailing efforts from international standards bodies, multilateral organizations, and national governments. It then details the importance of human-centered AI governance achieved through participatory mechanisms and examines various mechanisms for evaluating AI algorithms, which is essential to ensuring responsible deployment of systems, measuring harms, and shaping robust policies.

3.1 Regulatory Frameworks and Standards

The landscape of AI governance is shaped by a complex interplay of international standards, ethical guidelines, and regulatory frameworks. These structures operate at multiple levels, from technical specifications to broad ethical principles, creating a layered approach to algorithmic oversight.

International AI Standards

International standards bodies have developed frameworks that address both technical and governance aspects of AI systems. The ISO/IEC JTC 1/SC 42 (https://www.iso.org/committee/6794475.html), a subcommittee of the International

Organization for Standardization (ISO), has developed multiple technical specifications for AI implementation, formally publishing 37 standards on AI with 47 more in development as of August 2025. The published standards can be classified into three categories: setting best practices for AI development, providing an overview of AI and its respective risks, and providing guidelines on data processing and usage. For example, the ISO/IEC TS 4213:2022 (https://www.iso.org/standard/79799.html) standard specifies methodologies for measuring the classification performance of machine learning models, systems, and algorithms, the ISO/IEC TR 24028:2020 (https://www.iso.org/standard/77608.html) standard provides an overview of trustworthiness in AI, and standard ISO/IEC 24668:2022 (https://www.iso.org/standard/78368.html) provides a framework for organizations to leverage big data analytics. While ISO/IEC 38507:2022 (https://www.iso.org/standard/56641.html) is the only published standard from the ISO that explicitly focuses on governance, these standards published by ISO/IEC JTC 1/SC 42 overall can help create a common language for discussing the governance of AI-based algorithms across jurisdictions and sectors, facilitating international collaboration and system interoperability. The Institute of Electrical and Electronics Engineers Standards Organization (IEEE SA) has also developed over 120 standards through its initiative on Autonomous and Intelligent Systems (AIS) (https://standards.ieee.org/initiatives/autonomous-intelligence-systems/). Three of these standards focus specifically on algorithms, including IEEE 2841™-2022, which focuses on improving algorithm reliability, IEEE P2976™, which sets requirements for algorithmic explainability, and IEEE 7003™, which outlines methodologies to mitigate algorithmic bias.

Overall, mechanisms such as technical standards can be leveraged to establish governance frameworks for managing AI systems throughout their lifecycle. While international standards offer a foundation for AI governance through technical specifications and ethical frameworks, their efficacy remains limited by several critical factors. The voluntary nature of these standards creates uneven implementation across jurisdictions, while rapid technological advancement often outpaces the deliberate standardization process, creating persistent governance gaps. Moreover, these frameworks risk embedding Western-centric perspectives that may not adequately address diverse cultural contexts or the needs of developing nations, potentially exacerbating global digital divides rather than mitigating them. Additionally, without robust enforcement mechanisms or legal accountability, standards alone cannot address the power asymmetries between tech companies and regulatory bodies, raising questions about whether such voluntary governance frameworks genuinely serve public interest or merely provide the appearance of responsible oversight while enabling continued technological proliferation with minimal constraints.

Ethical AI Guidelines and Frameworks

Global organizations and governments have developed a number of ethical frameworks that provide foundational principles for AI governance. However, few of them specifically focus on AI-based algorithms, which is not unexpected given the complexity of designing and implementing policies for discrete aspects of AI systems. The Complex Adaptive System Framework to Regulate Artificial Intelligence, developed by the Indian Economic Advisory Council to the Prime Minister (EAC-PM), proposes several efforts for governing AI-enabled algorithms, which includes developing a national registry of algorithms, openly licensing algorithms to enable external audits, and setting standards for algorithmic explainability (Sanyal et al., 2024). The Council of Europe Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law is the first legally binding treaty to govern activities within the lifecycle of AI systems (Council of Europe, 2024). It requires developers to document relevant information regarding AI systems, enables consumers to challenge decisions from AI systems, and provides rights to those negatively affected by AI-based decision-making. The UN Resolution on AI, passed at the 2024 United Nations General Assembly meeting, emphasizes the need for member states to respect human rights and international law in AI deployment (United Nations, 2024). While not specific to the underlying algorithms of AI systems, this resolution can help guide the development of AI-based algorithms that do not infringe on human rights. The OECD AI Principles, adopted in 2019 and updated in 2024, incorporate five values-based principles to establish guidelines for responsible stewardship of trustworthy AI, focusing on sustainable development, human rights, transparency, security, and accountability. Like the other frameworks mentioned in this section, these principles do not specifically cover AI-based algorithms but can be easily applied to help steer responsible development.

While comprehensive in outlining normative principles, ethical AI frameworks, like international AI standards, suffer from critical limitations that undermine their practical impact. Most notably, they remain largely aspirational rather than operational, lacking specific implementation guidance and enforcement mechanisms that would translate lofty ideals into tangible algorithmic safeguards. The frameworks also exhibit a problematic tendency toward Western ethical paradigms that may not resonate globally, potentially creating ethical imperialism rather than genuine cross-cultural consensus. Additionally, these guidelines often address AI systems holistically without engaging with the technical nuances of algorithmic design where ethical failures frequently originate, creating a disconnect between high-level principles and ground-level development practices. Perhaps most concerning is how these frameworks risk becoming performative exercises that provide ethical cover for organizations while allowing harmful algorithmic practices to continue unabated beneath a veneer of compliance, raising questions about whether such voluntary governance approaches can effectively address the power asymmetries inherent in AI development and deployment.

Government-Led AI Regulation

National and regional governments have begun implementing specific AI regulations that reflect their particular societal values and priorities. These regulatory approaches range from comprehensive frameworks to sector-specific guidelines, creating a diverse regulatory landscape that AI systems must navigate. The impact of these regulations builds upon compliance requirements to influence AI system design, development practices, and deployment strategies. The General Data

Protection Regulation (GDPR), adopted by the European Union in 2016, has emerged as an influential model for data protection and algorithmic governance, introducing concepts like privacy by design and the right to explanation. GDPR specifies regulatory guidelines for AI systems more generally and, throughout the formal document, provides specific mentions for AI-based algorithms. For example, the Act requires organizations to provide technical documentation detailing the design specifications of algorithms, privacy measures when developing algorithms, and cybersecurity protection for AI-based algorithms, along with other components of AI systems. Brazil's proposed AI Act, which was approved by the Brazilian Senate in December 2024, requires organizations to conduct algorithmic impacts for systems classified as high risk and publicly disclose the results in alignment with trade secrecy requirements (Zanatta & Rielli, 2024). The Chinese government has introduced some of the most extensive set of regulations related to AI algorithms, primarily through the enacted Provisions on the Management of Algorithmic Recommendations in Internet Information Services (Cyberspace Administration of China, 2021). This law requires organizations leveraging recommendation algorithms to register their algorithms within the Internet Information Service Algorithm Filing System, along with other requirements that mandate measures to assess algorithmic mechanisms, strengthen cybersecurity practices, and prevent the use of algorithmic recommendation services for illegal activities. Other measures such as Provisions on Management of Deep Synthesis in Internet Information Service require review and verification of algorithmic mechanisms (Cyberspace Administration of China, 2022) and the Interim Measures for the Management of Generative Artificial Intelligence Services requires measures to prevent algorithmic discrimination, protection of intellectual property rights, promotes independent innovation, mandates filing, and necessitates cooperation with oversight inspections (Cyberspace Administration of China, 2023). In lieu of federal AI regulation in the United States, individual states within the country have drafted 1143 specific provisions for AI-based algorithms as of August 2025, covering domains such as criminal justice, healthcare, housing, social media, insurance, and finance (NCSL, 2025). Given that 183/1143 of these measures have been enacted or adopted, with the majority pending review, vetoed, or failed, there is much room to improve the governance landscape for specific algorithmic functions to ensure that AI governance is effective and as robust as possible.

3.2 Human-Centered AI Governance Approaches

To enable the responsible development of governance approaches for AI and AI-based algorithms, these processes should be participatory and leverage human-centered methods such as public consultations, multistakeholder convenings, and advisory boards. These approaches can ensure that the resulting frameworks and regulations are inclusive of diverse perspectives and needs while being robust enough to provide sufficient protections for end users and consumers.

Stakeholder Involvement

Methodologies for human-centered AI research and development often advocate for meaningful engagement and inclusion with stakeholders across the AI research and development lifecycle (Shneiderman, 2022; Xu & Gao, 2024). Similarly, the effective governance of AI systems also requires meaningful engagement with diverse stakeholder groups throughout the process of conceptualizing, drafting, and enacting governance frameworks and regulations (Png, 2022; Moon, 2023; Sigfrids et al., 2023). This inclusive approach brings together a number of stakeholders: developers who understand technical constraints and possibilities, policymakers who grasp regulatory requirements and public interest, users who experience the direct impacts of AI systems, civil society organizations that represent broader societal interests, and domain experts who understand context-specific implications. This multistakeholder approach ensures that governance frameworks reflect diverse perspectives and address real-world concerns while remaining technically feasible. A number of governance efforts, such as the EU AI Act, the AU-AI Continental Strategy, and UNSECO's Consultation Paper on AI Regulation, have leveraged multistakeholder consultation processes to solicit public feedback, identify aspects to improve, and crowdsource essential information (European AI Office, 2024; African Union, 2024; UNESCO, 2024). These consultative processes take varying forms and lengths; for example, the EU held its respective consultation process for 4 weeks from November to December 2024, and contributors were asked to provide their feedback through a form. Similar to the EU, UNESCO held a consultory period of 1 month from August to September 2024, soliciting feedback through a form. Along with leveraging forms to solicit feedback, the African Union Commission held online multistakeholder consultations that included panel discussions, question-and-answer (Q&A) periods, and a reflection session. While there is still a need for greater transparency in how stakeholder feedback from consultative processes is incorporated into final documents, these respective processes highlight the value that public consultations can provide in shaping more robust AI governance mechanisms.

Participatory AI Governance Models

Community-driven AI oversight has emerged as a promising approach to ensuring local context and values are respected in the development and implementation of AI governance. Such oversight is part of a broader process that can be classified as "participatory governance," where impacted stakeholders and end users are incorporated into the design, development, and implementation of regulatory mechanisms (Wong et al., 2022; Parthasarathy et al., 2024; Papagiannidis et al., 2025). For the governance of AI-based algorithms, these models range from citizen advisory boards like the United States National AI Advisory Committee (NIST, 2021) and the United Nations High-level Advisory Body on Artificial Intelligence (UN ODET, 2023) to community-led auditing initiatives, which have been employed decades before AI-based algorithms became embedded in decision-making systems (Vecchione et al., 2021). Within the United States, a number of state governments, such as Georgia (https://gta.georgia.gov/policies-and-programs/artificial-intelligence/ai-advisory-council), Tennessee (https://www.tn.gov/finance/ai-council.html), Vermont (https://governor.vermont.gov/Board_Commission/Artificial_Intelligence), Pennsylvania (https://www.legis.state.pa.us/CFDOCS/billInfo/billInfo.cfm?syear=2023&sInd=0&body=S&type=R&bn=143), Texas (https://dir.texas.gov/news/artificial-intelligence-advisory-council-appointed-study-use-ai-state-government), and Oregon (https://www.oregon.gov/eis/pages/ai-advisory-council.aspx), have instated AI Advisory Councils to guide state-level government adoption of AI, advise state government development of AI technologies, and shape public educational outreach efforts. Such participatory approaches help identify potential harms and benefits that might be overlooked in traditional governance frameworks while building public trust and understanding of AI systems. Participatory governance models can be especially useful in helping shape the development of AI-based algorithms, more specifically, to ensure that they operate in alignment with end user needs. However, companies and other institutions must be aware of privacy concerns when undergoing external oversight, along with ensuring they can protect their respective intellectual property when engaging in technical audits and evaluations.

3.3 Mechanisms for Evaluating AI Algorithms

Algorithmic Auditing

Algorithmic auditing serves as a crucial mechanism for evaluating AI system behavior and impacts (Raji et al., 2020; Metaxa et al., 2021). These methods, which have been classified into three levels by the Digital Regulation Cooperation Forum, include governance, empirical, and technical audits (DRCF, 2022). Governance audits leverage methods like impact assessments and compliance audits to understand how well governance policies have been followed in deploying algorithmic systems. Empirical audits aim to measure the impacts of algorithmic systems as they are implemented and used in real-world contexts. These audits are primarily focused on the outcomes of AI systems and often employ quantitative methods to measure harm without having direct access to the underlying algorithm embedded within the respective AI system. Technical audits, which are becoming a popular method for evaluating the capabilities and risks of AI systems and their algorithms, focus on reviewing the data sources used to train AI algorithms, the source code of AI models, and methods used to design and implement AI systems. To help formalize algorithmic auditing approaches, a number of governments have established AI Safety Institutes to guide governmental evaluation of AI algorithms and responsible implementation of systems (Fort, 2024; Araujo et al., 2024). These institutes have established comprehensive requirements for audits, including technical performance evaluation across diverse scenarios, assessment of bias and fairness metrics, security vulnerability testing, privacy impact analysis, robustness testing under various conditions, and documentation review and verification. These requirements provide a structured approach to identifying and addressing potential issues before they manifest in deployed systems. However, there are increasing concerns about the limited effectiveness of algorithmic audits (Brown et al., 2021; Costanza-Chock et al., 2022; Urman et al., 2024), which can lead to a phenomenon known as "audit washing," where subpar algorithmic auditing practices provide an illusion of that AI algorithms and systems are operating reliably and safely (Goodman & Tréhu, 2022).

Algorithmic Impact Assessments

Algorithmic impact assessments provide a systematic way to evaluate the broader societal implications of AI-based algorithms embedded within AI systems before these tools are implemented for real-world use (Metcalf et al., 2021). These assessments are usually conducted by developers of AI systems, organizations procuring these tools, and external auditors who are contracted on behalf of the procuring organizations. This provides stakeholders across the AI value chain, including developers and the general public, with an opportunity to preview concerns from algorithmic systems before formal contracting takes place and ahead of widescale deployment (Reisman et al., 2018). Algorithmic impact assessment methods build upon existing approaches to impact assessments that have been deployed in domains such as environmental protection (Morgan, 2012), human rights (Tedaldi et al., 2016), and healthcare (Kemm et al., 2004). Algorithmic impact assessments examine direct and indirect effects on various stakeholder groups, potential social and economic impacts, environmental considerations, human rights implications, and long-term societal consequences. A number of governments and organizations have developed frameworks and formal processes to guide algorithmic impact assessments, such as the Canadian government's Algorithmic Impact Assessment Tool (https://www.canada.ca/en/government/system/digital-government/digital-government-innovations/responsible-use-ai/algorithmic-impact-assessment.html) and the Ada Lovelace Institute's Algorithmic Impact Assessment User Guide (Ada Lovelace Institute, 2022). Engaging in such comprehensive evaluation can help identify potential risks and benefits early in the development process, enabling proactive governance approaches. However, similar to algorithmic auditing, algorithmic impact assessments can be prone to shortcomings due to a lack of inclusion in the assessment process (Kneese, 2024), insufficient governance structures (Selbst, 2021), and deficient standardization (Ashar et al., 2024). With this in mind, algorithmic impact assessments must be crafted carefully to ensure that they are domain-specific, provide clear definitions of the range of potential impacts (including how harmful, positive, or neutral impacts are defined), and incorporate interpretable metrics that enable stakeholders to effectively mitigate harm from AI-based algorithmic systems.

Algorithmic Transparency Tools

Tools for promoting algorithmic transparency can operate at multiple levels. They could include technical tools that provide insight into model behavior and decision-making processes, documentation requirements that ensure clear communication about system capabilities and limitations, visualization tools that make algorithmic processes more comprehensible to nontechnical stakeholders, or reporting frameworks that standardize how information about AI systems is communicated. Popular examples of algorithmic transparency tools include model cards, an approach developed by researchers at Google that consists of short documents that enable machine learning model developers to provide information on the details of models, their intended use, a summary of model performance across various factors, model metrics, data used to evaluate and train the model, results of quantitative analyses, ethical considerations, and additional concerns that should be communicated about the model (Mitchell et al., 2019). To help enhance algorithmic transparency, a number of efforts have focused on refining approaches like model cards (Crisan et al., 2022; Boone et al., 2022) to improve accessibility and user comprehension, both of which are key aspects of human-centered AI. To enable best practices for developing AI-based algorithms, it is also important to set standards for the data used to train these respective models. Popular methods focusing on dataset development and documentation include Dataset Nutrition Label, which assesses the quality of datasets through their collection, composition, and management (Holland et al., 2020,) and Datasheets for Datasets, which provides documents that detail how datasets were created, their respective composition, intended uses, maintenance structures, and legal and ethical considerations (Gebru et al., 2021). These tools support meaningful oversight by making algorithmic systems more understandable and accountable to various stakeholders.

4 Human-Centered Governance Process of AI-Based Algorithms

The diverse governance approaches explored thus far underscore the need for a more focused, human-centered approach to governing AI-based algorithms, ensuring that ethical principles are translated into practical processes. Therefore, shifting from overarching strategies to specific implementation, it is vital to establish goals that prioritize human values, adhere to regulations, manage risks, leverage transparency, and ensure accountability and redress. This section emphasizes the necessity of adapting AI systems to legal standards, institutionalizing regulatory compliance within organizational governance structures, promotes dynamic, proactive approaches to risk management, highlights the crucial role of comprehensive documentation and interpretable AI models in facilitating transparency, and stresses the critical need for establishing clear accountability frameworks and accessible redress mechanisms.

4.1 Defining Human-Centered Goals and Objectives

The foundation of human-centered AI governance lies in establishing clear objectives that prioritize human values and interests throughout the algorithmic lifecycle. This approach recognizes that while AI-based algorithms and the systems they are embedded in offer powerful capabilities, their development and deployment must fundamentally serve human needs and respect human agency. Human-centered goals in AI governance must broaden traditional performance metrics to encompass broader societal considerations, including the preservation of human autonomy, the promotion of individual and collective well-being, and the protection of fundamental rights (Sigfrids et al., 2023; Gentelet & Mizrahi, 2023). These objectives must be operationalized through concrete governance frameworks that balance technological advancement with human values. Additionally, these respective frameworks must be contextualized to the specific domains they will be applied in, along with accounting for differences in the governing capacities of enforcement agencies and offices. This includes developing specific criteria for evaluating the impact of AI-based algorithms on human autonomy, establishing clear guidelines for human oversight and intervention, and creating mechanisms to ensure that AI development aligns with societal values and ethical principles. The process requires ongoing dialogue between diverse stakeholder groups such as technologists, policymakers, civil society, and the general public to ensure that governance objectives remain responsive to evolving societal needs and concerns.

4.2 Regulatory Compliance and Legal Considerations

The integration of AI systems into various sectors necessitates careful attention to regulatory compliance and legal frameworks. In sectors like healthcare, education, and finance, where sensitive information is more likely to be protected through domain-specific data privacy regulation (Soloway & Covington, 2006; Ritvo et al., 2013; Dove & Phillips, 2015), regulatory measures focused on the data used to train AI-based algorithms must work in tandem with existing policies rather than trying to override them. Contemporary AI governance must navigate a complex landscape of existing and emerging regulations, from established data protection frameworks to newer AI-specific legislation. This regulatory environment shapes the development and deployment of AI-based algorithms and their respective systems, establishing baseline requirements for data handling, algorithmic transparency, and system accountability. The evolving nature and capabilities of AI-based algorithms require governance frameworks that can adapt to these respective technical capacities. These frameworks must be flexible enough to account for new regulatory requirements while maintaining consistent human-centered principles. This involves developing robust compliance mechanisms that can accommodate varying jurisdictional requirements, implementing systematic approaches to regulatory tracking and updates, and establishing clear protocols for adapting AI systems to meet new legal standards as they emerge. As regulatory measures that are specific to AI gradually enter into force, organizations must also work to institutionalize regulatory compliance into their corporate governance structures and train sufficient personnel to manage and supervise implementation processes (Schultz & Seele, 2023).

4.3 Risk Management and Monitoring

Effective risk management in AI governance requires a dynamic approach that combines proactive risk assessment with continuous monitoring capabilities. This process involves identifying potential risks across technical, social, and ethical dimensions, implementing mitigation strategies, and establishing monitoring systems that can detect and respond to emerging issues in real time. The most prominent framework for managing risks within technical systems is the U.S. National Institute for Standards and Technology (NIST) Risk Management Framework, which defines a process for integrating guidelines for cybersecurity, privacy, and incident response (NIST, 2016). This framework has been further refined for AI systems to promote trustworthy design, development, usage, and evaluation of AI systems and tools (NIST, 2023). The complexity of AI-based algorithms demands sophisticated monitoring approaches that can track both direct system outputs and broader societal impacts. With this in mind, organizations can rely on frameworks such as NIST RMF but must also work to develop their own contextualized frameworks to fit their respective scope and scale of AI development, procurement, adoption, and implementation. Risk management frameworks must address various types of risk, from technical failures and security vulnerabilities to ethical concerns and societal implications. This requires developing comprehensive risk assessment methodologies, establishing clear risk thresholds and triggers for intervention, and implementing monitoring systems that can track system performance across multiple dimensions. Additionally, frameworks should enable quick detection of anomalies and provide clear protocols for risk mitigation and system adjustment when necessary.

4.4 Leveraging Transparency and Interpretability

Comprehensive documentation of stages within the AI lifecycle serves as a cornerstone of transparency and can help facilitate human-centered AI governance (Königstorfer & Thalmann, 2022; Winecoff & Bogen, 2024). Documentation includes maintaining detailed records of model design choices, training data characteristics, evaluation metrics, and system modifications. Documentation must capture not only technical specifications but also the reasoning behind key decisions and the potential implications of various design choices. Many of these respective methods were discussed in the section "Mechanisms for Evaluating AI Algorithms" and provide solid starting points for organizations and individuals developing, procuring, and implementing AI systems. Transparent algorithmic development processes can also play a significant role in developing interpretable AI models. Interpretable AI represents a crucial aspect of human-centered AI governance, especially as it pertains to communicating the functionality of AI models, constraints of AI systems, and reasoning behind the outputs of decisions produced from AI-based algorithms (Danks, 2022). Given the complexity of AI-based algorithms and the varying functions of AI systems, developing robust and effective interpretability approaches is still an emerging area of future inquiry (Hong et al., 2020;

Krishnan, 2020; Watson, 2022). The section "Transparency and Explainability" in this chapter also focuses on the relationship between transparency and explainable AI. Interpretable AI is a complement to explainable AI, involving the creation of algorithms that not only perform effectively but also provide clear insights into their decision-making processes. Interpretability must be considered from the earliest stages of model development, influencing choices in model architecture, feature selection, and training methodologies.

4.5 Accountability and Redress

Establishing clear accountability frameworks is essential for ensuring that AI-based algorithms are developed and implemented responsibly (Durante & Floridi, 2022). This includes defining specific roles and responsibilities for various stakeholders, implementing mechanisms for tracking and attributing decisions, and creating clear channels for addressing concerns and grievances. Accountability frameworks must be supported by concrete mechanisms for redress, enabling affected individuals or groups to seek remediation when AI systems cause harm or make incorrect decisions (Ogunleye, 2022; Fanni et al., 2023). Redress mechanisms should be accessible, efficient, and proportionate to the potential harms they address. This includes establishing clear procedures for filing complaints, conducting rigorous investigations, and implementing corrective measures to prevent harm from reoccurring. Additionally, redress frameworks should include preventive measures that learn from past incidents to improve system performance and prevent similar issues in the future. This human-centered approach to AI governance emphasizes the importance of maintaining human oversight and control by acknowledging the risks of AI in order to ensure that the benefits of these technologies can be effectively leveraged. Within human-centered governance, incorporating principles of accountability and redress enables stakeholders to recognize that effective governance requires ongoing attention to human needs, values, and rights throughout the AI lifecycle, supported by robust mechanisms for ensuring compliance, managing risks, and addressing concerns as they arise.

5 Challenges and Ethical Implications of Governing AI-Based Algorithms

While a robust human-centered governance process is essential, it is equally important to acknowledge and address the various challenges and ethical implications that arise in practice. Beyond establishing abstract principles, meaningful governance of AI-based algorithms demands directly confronting the practical challenges of implementation, as these represent the actual terrain where principles either materialize into substantive protections or deteriorate into performative ethics that legitimize rather than constrain harmful algorithmic practices. This section will cover various issues, including the ability for regulatory approaches to maintain pace with rapid AI innovation, regulatory enforcement gaps across jurisdictions, methodological limitations in algorithmic evaluation, balancing human autonomy with AI decision-making, and navigating the tradeoff between data privacy and system robustness.

5.1 Regulatory Pace with AI

The rapid evolution of AI technologies presents a fundamental challenge to traditional regulatory frameworks, which typically operate on much slower timescales (Fenwick et al., 2016; Hagemann et al., 2018). Because interest in developing and adopting AI solutions has also risen drastically over the past few years, there is insufficient information available on how these tools operate in a variety of real-world contexts outside of countries with high rates of AI adoption, providing a lack of empirical evidence that can help craft robust regulation. While AI capabilities advance exponentially, regulatory processes remain bound by deliberative procedures, public consultation requirements, and legislative cycles. This temporal mismatch creates governance gaps where emerging AI applications may operate without adequate oversight. These challenges extend beyond mere speed differentials and encompass the difficulty of crafting regulations that remain relevant and effective as underlying technologies continue to evolve. Additional regulatory gaps may arise, given that governments may be inclined to govern AI more broadly without making considerations for the specific functions of these technologies, their respective classifications (predictive, agentic, generative, etc.), and how these technologies function in real-world contexts. Moreover, the development of concrete AI policies may also be inhibited by national motivations to maintain competitiveness in AI. This competitive dynamic generates a "race to the bottom" effect where jurisdictions with less stringent regulations gain short-term technological advantages but potentially compromise long-term ethical safeguards and public trust. The development of AI-specific policies requires substantial time and resources and is further complicated by the need to harmonize regulations across jurisdictions and sectors. This process involves balancing competing interests, gathering expert input, and conducting impact assessments while ensuring policy coherence across different regulatory domains. Furthermore, the challenge of harmonization becomes particularly acute in global contexts, where varying national priorities and regulatory approaches must be reconciled to create effective governance frameworks. The trade-off between national competitiveness, functional considerations of AI-based algorithms, and effectively addressing ethical concerns will require innovative governance approaches that can resolve regulatory tensions without sacrificing either of these priorities.

5.2 Regulatory Compliance, Enforcement, and Reporting

Enforcing AI regulations can present unique challenges due to the complexity and opacity of AI-based algorithms and the systems they are embedded in

(Weissinger, 2024). Traditional compliance mechanisms may prove inadequate for assessing algorithmic behavior, particularly in systems that exhibit tasks they are not directly programmed for or undergo continual learning, which enables AI models to adaptively learn new tasks (Wang et al., 2024). To account for these nuances, enforcement bodies must develop new competencies and tools to effectively monitor compliance, often requiring sophisticated technical expertise that may be scarce in regulatory agencies. To aid compliance, human-centered AI governance mechanisms must also advocate for reporting requirements to gain clear information on model performance, usage, and failures. Reporting also aligns with the concept of establishing formal algorithmic and model documentation methodologies, which is discussed in the "Mechanisms for Evaluating AI Algorithms" section. Establishing meaningful reporting requirements for institutions developing and adopting AI demands careful consideration of what information is both relevant and feasible to collect. There are also additional challenges in defining reporting metrics that provide genuine insight into algorithmic performance and AI system behavior while not imposing unreasonable burdens on innovation (Goodman & Tréhu, 2022). Additionally, concerns that implementing audits and subsequent reporting measures may compromise intellectual property could potentially disincentivize organizations from implementing these approaches (Bell et al., 2023). To mitigate IP theft risks, organizations should focus on determining appropriate transparency levels, standardizing reporting formats, and ensuring that reported information enables meaningful oversight.

5.3 Development of AI Evaluation Measures

To evaluate the capabilities of AI models, assess potential risks, and understand algorithmic functionality, AI developers and researchers often rely on benchmarks. AI benchmarks are standardized tests that can be in text, image, or multimodal form and are used to measure the performance of AI systems, models, algorithms, or specific components within them. Overall, these methods provide formal ways to compare different AI approaches and track progress within the field (Tolan et al., 2021). The technical challenges in implementing robust measures for evaluating AI systems stem from the inherent complexity of algorithmic behavior and the difficulty of quantifying potential harms. These challenges are further exacerbated by relying on external evaluation frameworks, leveraging crowdsourced feedback in evaluations, and using generative AI models to generate evaluation benchmarks (Ganguli et al., 2023). Traditional evaluation metrics often fail to capture the full range of potential impacts, particularly in areas involving subtle social dynamics or long-term societal effects. Developing comprehensive evaluation frameworks requires addressing challenges in measuring fairness, bias, transparency, and system reliability across diverse contexts and applications. However, concepts like transparency vary in definition, have no established metrics, and can be measured differently across the AI value chain (Hayes et al., 2022), potentially increasing the difficulty for model developers to evaluate AI-based algorithms and models effectively.

Moreover, researchers have found issues with consistency in benchmark quality, lack of replication of results, and substandard reporting on evaluation metrics (Reuel-Lamparth et al., 2024), further necessitating improved algorithmic evaluations.

5.4 Human Autonomy and AI Decision-Making

The relationship between human oversight and allowing AI systems to operate autonomously, where there are constraints on human intervention, represents a central tension in governing AI-based algorithms. While there is still research needed to evaluate the benefits of developing and implementing autonomous AI systems, relying solely on this form of functionality may also increase the risk of harm and introduce accountability gaps (Sifakis, 2019). Additionally, there are concerns that relying too heavily on AI systems for decision-making and recommendations can undermine human autonomy (Laitinen & Sahlgren, 2021; Prunkl, 2024; Lu, 2024). While increasing interest in agentic AI indicates that autonomous AI systems may become more prominent, these systems are also prone to algorithmic harm, emphasizing the need for human oversight and intervention (Chan et al., 2023). Finding the appropriate balance between the implementation of autonomous AI systems and reducing negative impacts on human autonomy will require careful consideration of context-specific factors, including limiting autonomous AI for critical decisions, calculating the potential for harm, and providing mechanisms for meaningful human supervision. As discussed previously in the “Accountability and Redress” subsection, delegating decision-making authority to AI systems raises profound ethical questions, particularly in domains with significant human impact. In healthcare, decisions affect patient outcomes and well-being; in finance, they influence economic opportunities and resource allocation; in law enforcement, they impact individual liberty and public safety. These contexts and other sensitive use cases demand careful consideration of when and how decision-making authority should be shared between human stakeholders and artificial agents. Finally, power differentials in global AI governance structures, which currently benefit more dominant states in the AI “race” such as those in North America, the EU, and the United Kingdom, also raise implications for nations in the Global Majority to have sufficient autonomy in developing and governing AI systems (Anthony et al., 2024; Gwagwa & Mollema, 2024). Because these countries are heavily reliant on AI algorithms, models, and access to computing infrastructure from major tech companies and research institutions, they also tend to play a less significant role in shaping global AI governance initiatives. Thus, it is essential that efforts focused on mitigating the impacts of AI and human autonomy also be aware of these power imbalances and how they impact decision-making across varying global contexts.

5.5 Privacy and Surveillance

Challenges in implementing algorithmic governance are often grounded in frictions regarding complying with data privacy protections versus optimizing for AI system effectiveness (CIPL, 2018; OECD Secretariat, 2024). While AI systems often benefit from access to large datasets, this access must be balanced against individual privacy rights and data protection principles. The deployment of AI-powered systems exacerbates surveillance, challenging rights to privacy, fueling algorithmic bias, repressing free speech, and enabling corruption (Kalluri et al., 2023). AI systems specifically developed to aid surveillance, such as those built in the interest of "national security," like airport biometric systems and refugee identification systems, also pose risks to civil liberties and human rights (Farraj, 2010; Smith & Miller, 2021). As organizations look to refine their AI algorithms and governments increase investments in surveillance technologies, these respective interests end up creating trade-offs between individual privacy and organizations' needs to diversify and expand their respective data sources to train underlying algorithms. These challenges also prioritize the need for the development of governance frameworks that acknowledge the role that data plays in training, refining, and evaluating AI systems while preventing its misuse and protecting fundamental rights. Human-centered AI governance approaches can also inform the development of novel adaptive and nuanced approaches that can evolve alongside the technical capabilities of AI-based algorithms while maintaining strong protections for human values and rights (Sigfrids et al., 2023). Effectively implementing these approaches will require ongoing dialogue between stakeholders, continuous refinement of governance mechanisms, and careful attention to the sociotechnical dimensions shaping AI system deployment (Bogen & Winecoff, 2024).

6 Case Studies of AI-Based Algorithm Governance

The complexities outlined in the discussion of challenges and ethical implications are particularly salient in sectors where AI-based algorithms have a profound impact on individuals and society. Therefore, it is crucial to explore how these challenges materialize and are addressed within specific domains. This section explores the governance of AI-based algorithms in healthcare, where patient safety is paramount; finance, with its potential for exacerbating historical marginalization; and social media, which shapes public discourse and information flows. It emphasizes the necessity of understanding sector-specific nuances, implementing targeted governance mechanisms, and learning from diverse approaches to manage the ethical and practical implications of AI-based algorithms.

6.1 Healthcare

The regulation of healthcare algorithms is a multifaceted issue, particularly in the United States, where regulatory frameworks are still adapting to the pace of technological innovation. While many algorithms might be classified as medical devices requiring pre-market approval, others may be considered as services or products, leading to less stringent protections. Regulation on AI-based algorithms in healthcare primarily focuses on preventing bias, ensuring transparency, and guaranteeing patient safety, especially as AI-powered systems become increasingly common in healthcare decision-making. This includes measures to avoid algorithmic discrimination, require impact assessments for AI systems, and notify individuals of AI-generated decisions. There have been a growing number of regulatory efforts to govern AI-based algorithms in the healthcare sector, most notably from US government agencies like the Department of Health and Human Services (HHS) and the Food and Drug Administration (FDA).

In 2024, the Centers for Medicare & Medicaid Services within HHS issued the Nondiscrimination Final Rule as part of Section 1557 of the Affordable Care Act, prohibiting discrimination by health programs and activities using AI and other automated decision-making systems (HHS, 2024a). While the proposed rule initially included the phrase “clinical algorithms,” it was replaced in 2024 with the term “patient care decision support tool,” which is defined as any automated or non-automated tool, mechanism, method, technology, or combination thereof used by a covered entity to support clinical decision-making in its health programs or activities.” In the HTI-1 Final Rule, the Office of the National Coordinator for Health IT (ONC) mandates that developers of AI and predictive algorithms within certified health IT provide users with access to information about the algorithm’s purpose, development, performance, and fairness (HHS, 2024b). From November 1995 to September 2024, the FDA has approved 1016 medical devices that are embedded with AI-enabled algorithms, focusing on deterministic algorithms which consistently produce the same output for a given input and remain unchanged through use (FDA, 2024). These evolving regulatory frameworks represent incremental progress toward human-centered governance that recognizes the role of algorithms as sociotechnical interventions that require robust oversight. The healthcare sector thus serves as a critical testbed for governance approaches that balance innovation with protection, particularly in contexts where algorithmic errors can directly affect patient outcomes and exacerbate existing healthcare inequities.

6.2 Finance

The governance of AI-based algorithms in financial services represents a critical frontier where technological innovation intersects with systemic risk and individual rights, demanding regulatory frameworks that can address both technical and social dimensions of algorithmic power. Financial markets increasingly rely on algorithmic systems that operate at speeds and scales beyond human oversight capabilities, from high-frequency trading algorithms that execute thousands of transactions per second to automated credit scoring systems that determine financial access and opportunity for millions of individuals. Despite the absence of comprehensive global regulations specifically targeting AI in finance, implementing robust governance mechanisms is essential not merely as a regulatory compliance exercise but as a fundamental safeguard against market manipulation, systemic instability, discriminatory lending practices, and the amplification of existing socioeconomic inequalities. The stakes of algorithmic governance in finance are particularly high given that these systems mediate access to essential financial resources and services, making their operation a matter of both economic stability and distributive justice.

The National Association of Insurance Commissioners (NAIC) Model Bulletin on the Use of Artificial Intelligence Systems by Insurers, unanimously approved by NAIC members in December 2023, provides guidance to state insurance regulators regarding the responsible use of AI systems and the algorithms embedded within these systems by insurers (NAIC, 2023). It outlines expectations for insurers regarding the development, acquisition, and use of AI, emphasizing areas like governance, risk management, and potential biases. The Bulletin aims to ensure AI systems are used ethically, safely, and in compliance with existing laws and regulations, particularly those related to unfair trade practices. As of August 2025, the NAIC Model Bulletin has been adopted in 24 jurisdictions across the United States, which includes Alaska, Arkansas, Connecticut, Delaware, District of Columbia, Illinois, Iowa, Kentucky, Maryland, Massachusetts, Michigan, Nebraska, Nevada, New Hampshire, New Jersey, North Carolina, Oklahoma, Pennsylvania, Rhode Island, Vermont, Virginia, Washington, Wisconsin, and West Virginia. Colorado is the first state to pass formal regulation for life insurers using external consumer data (ECDIS) and algorithms/predictive models (Colorado Division of Insurance, 2023). This regulation aims to prevent unfair discrimination based on race or other protected characteristics in insurance practices and requires insurers to establish a risk-based governance structure and management framework overseen by the board. Other states like California (https://www.insurance.ca.gov/0400-news/0100-press-releases/2022/release048-2022.cfm), Connecticut (https://portal.ct.gov/-/media/cid/1_notices/technologie-and-big-data-use-notice.pdf), New York (https://www.dfs.ny.gov/industry-guidance/circular-letters/cl2024-07), and Texas (https://www.tdi.texas.gov/bulletins/2020/B-0036-20.html) have issued formal bulletins on the use of algorithms in marketing insurance products and in practices such as insurance underwriting, rate pricing, credit rating, and claims processing. These state-level governance initiatives reveal the emergence of a federated regulatory approach that acknowledges the socially embedded nature of financial algorithms, while simultaneously highlighting the critical need for cross-jurisdictional harmonization to prevent regulatory arbitrage in which algorithmic systems migrate to less-regulated environments, ultimately suggesting that effective governance of AI-based algorithms in finance requires multistakeholder coordination across public and private sectors to ensure that technological innovation enhances rather than undermines financial inclusion and equity.

6.3 Social Media

Governing AI-enabled algorithms in social media systems serves as a critical frontier for democratic participation and information integrity in an increasingly digitally mediated public sphere. Social media algorithms, including recommendation engines that curate personalized content feeds, automated moderation systems that determine acceptable content, and engagement optimization algorithms that shape user behavior, now function as de facto gatekeepers of public discourse, wielding unprecedented influence over collective attention, belief formation, and social cohesion. Despite fragmented regulatory frameworks and insufficient industry self-regulation characterized by opacity and inconsistent enforcement, implementing robust governance mechanisms for these algorithmic systems is imperative not merely as a matter of consumer protection but as a fundamental safeguard for democratic processes, cognitive autonomy, and social trust. The societal stakes of algorithmic governance in social media are particularly significant given how these systems increasingly mediate core aspects of civic participation and knowledge formation, making their operation a matter of public interest that transcends traditional regulatory categorizations and demands innovative governance approaches capable of addressing both their technical operations and social consequences.

Regulatory approaches to AI-powered social media algorithms remain fragmented and geographically inconsistent, creating governance gaps despite growing recognition of their societal impact. While the United States currently lacks comprehensive federal data privacy regulation and AI governance frameworks, state governments have begun to legislate independently in the domain of social media. In the United States, New York State has passed the Stop Addictive Feeds Exploitation (SAFE) for Kids Act and the Child Data Protection Act, which is the first set of legislation limiting how social media companies leverage AI-based algorithms to target content toward minor users, placing restrictions on push notifications delivered overnight for algorithmic feeds, and limiting the data used to create algorithmic feeds and targeted advertisements (New York State Senate, 2024). Other regions like the EU have developed the Digital Services Act (DSA) to create a safer and more open online environment by addressing illegal content, transparent advertising, and disinformation (European Parliament, 2022). The DSA, adopted in 2022, also includes specific provisions for transparency regarding algorithms used for content moderation, ranking, decision-making, and recommendations. The United Kingdom's Online Safety Act passed in 2023, includes specific provision on AI-based algorithms, requiring social media providers to consider how their algorithms expose users to illegal content (UK DSIT, 2025). However, recent efforts toward governing AI-based algorithms used on digital platforms like Australia's proposed changes to their Online Safety Act that would have required social media platforms to limit algorithmic promotion of harmful content (Parliament of Australia, 2025) and proposed changes to the Digital India Act mandating accountability from social media platforms regarding personalized content recommendation (Aryan, 2022) have failed to pass or make significant progress. Strengthening governance of AI-based algorithms within the social media domain requires a multifaceted approach that integrates co-regulatory mechanisms where industry standards complement legislative guardrails, establishes independent oversight bodies with sufficient technical expertise and enforcement authority, and develops algorithmic impact assessment frameworks that evaluate not just technical performance but broader sociopolitical consequences. This human-centered governance approach must ultimately shift incentive structures of digital platforms away from maximizing engagement toward public interest metrics that prioritize information integrity and user autonomy over profit-driven motives.

7 Future Directions in AI-Based Algorithm Governance

The preceding case studies illuminate the real-world complexities and sector-specific challenges of governing AI-based algorithms. These examples demonstrate the growing patchwork of localized and domain-specific regulations, alongside varying approaches to implementation and enforcement. As AI technologies continue to advance at a rapid pace, there is an increasing urgency to move from soft law and aspirational principles to more formalized, implementable standards that can be effectively applied across borders. Achieving harmonized global governance will require navigating diverse national priorities, bridging varying regulatory capacities, and fostering equitable participation from all nations, particularly those in the Global Majority. This final section will explore these critical future directions, examining the evolving landscape of AI governance and the pathways toward effective international cooperation.

7.1 Emerging Trends in AI Governance

The landscape of AI governance is experiencing rapid evolution, shaped by technological advances and growing societal awareness of AI's impact. This evolution is marked by several key developments that are likely to define the future of algorithmic oversight and control. The regulatory environment for AI is becoming increasingly sophisticated, moving beyond broad principles toward specific, implementable standards. However, a vast majority of these frameworks, declarations, international statements, principles, multilateral commitments, and executive orders constitute soft law, informal governance mechanisms that are unenforceable by regulatory bodies (Hagemann et al., 2018; Marchant, 2019; Villasenor, 2020). Transforming current AI governance approaches will require the emergence of risk-based regulatory frameworks that leverage formal oversight mechanisms to mitigate the potential impact of AI systems and provide systematic avenues for redress. Such frameworks should recognize that AI applications applied to various domains require varying levels of scrutiny and control, allowing for more nuanced and effective governance approaches.

The standard-setting process is also evolving, with greater emphasis on practical implementation guidelines rather than abstract principles (Cihon, 2019; Auld et al., 2022). This shift reflects a growing understanding that effective AI governance requires concrete, actionable standards that can guide development and deployment decisions. These emerging standards increasingly incorporate metrics for measuring compliance and effectiveness, enabling more objective assessments of AI systems against governance requirements. The private sector's role in AI governance is also expanding through various self-regulatory initiatives (Heikkilä, 2024) and growing influence on the development of national AI strategies and action plans (Walter, 2024; FMCIDE, 2024). Additionally, AI ethics boards within companies are becoming more common and more influential, surpassing advisory roles to have more tangible impacts on development decisions (Cihon et al., 2021; Schuett et al., 2024). These boards increasingly include diverse stakeholders, from technical experts to ethicists and civil society advocates, reflecting a broad understanding of AI's societal implications.

Regulatory sandboxes have emerged as important tools for testing technological innovations in controlled environments with time-limited regulatory waivers (Truby et al., 2022; OECD Secretariat, 2023). Many approaches to AI regulatory sandboxing have been adopted from fields like financial technology, where the first regulatory sandbox in this sector was launched by the United Kingdom Financial Conduct Authority in 2015 (Cornelli et al., 2020). These experimental spaces allow organizations to trial new AI applications and governance mechanisms while managing risks and gathering evidence about their effectiveness. AI regulatory sandboxes are commonly implemented over a period of 6 months and involve stakeholders such as companies developing AI products, regulators, and consumers. This approach enables innovation while maintaining oversight, providing valuable insights for future regulatory development. In 2022, the European Union debuted the world's first regulatory sandbox on AI to help steer the implementation of the EU AI Act, which entered into force in 2024 (European Union Commission, 2022). While the EU AI Act still remains the world's only formal AI regulation, the continued development of AI regulatory sandboxes will be crucial as countries around the world scale their respective AI governance efforts.

7.2 Toward Global Governance of AI

The inherently global nature of AI technology necessitates international cooperation in governance efforts. This cooperation is becoming more structured and institutionalized, moving beyond informal agreements toward more binding frameworks. The future of AI governance will likely see the emergence of international bodies with specific mandates for overseeing algorithmic systems, similar to existing structures for financial or nuclear oversight (Neuwirth, 2024). However, discourse and efforts on international AI cooperation are predominantly driven by higher-income countries that are also at the forefront of AI development (Png, 2022; Okolo, 2025). To ensure that AI governance approaches work fairly and effectively for marginalized communities and lower-income regions, countries in the Global Majority must also be equitably included in these efforts. To work toward global AI governance, several key areas are emerging as focal points for international cooperation, including cross-border data flows and algorithmic decision-making, shared standards for AI safety and reliability, coordinated approaches to emerging AI capabilities, joint mechanisms for addressing global AI risks, and collaborative research on context-specific governance approaches.

The path toward harmonized global AI governance faces significant challenges but also presents opportunities for meaningful progress. The primary challenges stem from differing national priorities, varying capacities for regulatory enforcement, and competing economic interests (Kiani & Shafiee, 2022; Niazi, 2023). However, these challenges are balanced by a growing recognition of shared risks and opportunities in AI development. Governments and multilateral organizations interested in collaborating on global AI governance mechanisms should recognize key harmonization challenges and work to reconcile different cultural and ethical perspectives on AI governance, balance national sovereignty with global oversight needs, address economic disparities in AI development capabilities, manage competing regulatory approaches and traditions, and ensure equitable participation in governance development (Ala-Pietilä & Smuha, 2021; Roberts et al., 2024). Additionally, stakeholders in international AI cooperation must take advantage of emerging opportunities for harmonization, including developing shared technical standards and evaluation metrics, creating international AI incident reporting systems, establishing cross-border cooperation mechanisms, forming global AI research and governance networks, and implementing coordinated risk assessment frameworks (Tallberg et al., 2023).

The future of AI governance will likely be characterized by a hybrid approach that combines global standards with local adaptation (World Bank Group, 2024). This approach recognizes that while certain aspects of AI governance require universal standards, implementation must be sensitive to local contexts and needs. The success of frameworks leveraging this approach will be dependent on factors that prioritize flexible governance mechanisms that can adapt to technological change, the establishment of clear protocols for international cooperation and conflict resolution, balanced representation of diverse stakeholder interests, practical implementation guidelines that work across different contexts, and effective mechanisms for knowledge sharing and capacity building. As AI technology continues to advance, governance frameworks must evolve to address new challenges while maintaining core principles of human-centered development. The future effectiveness of AI governance will depend on the ability to balance innovation with protection, global standards with local needs, and technological capability with ethical considerations. This evolution requires ongoing dialogue between stakeholders, continuous refinement of governance mechanisms, and sustained commitment to international cooperation.

8 Conclusion

The governance of AI-based algorithms necessitates a comprehensive framework grounded in key principles, including transparency, accountability, explainability, fairness, privacy, and safety. These principles serve as the foundation for ethical and responsible AI development, ensuring that AI systems operate in ways that align with societal values and human rights. Transparency and explainability are vital for building trust in AI systems, while accountability mechanisms help assign responsibility and establish redress pathways. Addressing bias and ensuring fairness remain critical to preventing discriminatory outcomes, particularly in domains such as hiring, criminal justice, and finance. Privacy protections, shaped by regulatory frameworks like the GDPR, help safeguard personal data, while safety and reliability measures ensure that AI systems perform as intended, particularly in high-risk sectors.

Despite these guiding principles, several challenges persist in governing AI-based algorithms. The rapid pace of AI advancements often outstrips regulatory responses, creating governance gaps. Issues of bias, underrepresentation, and privacy concerns remain complex and multifaceted, requiring continuous refinement of regulatory frameworks. The question of liability remains particularly intricate, given the distributed nature of AI development and deployment. However, these challenges also present opportunities. The development of human-centered AI governance approaches, regulatory sandboxes, algorithmic impact assessments, and participatory governance models enables more inclusive and adaptive regulatory mechanisms. Collaborative international efforts to establish harmonized AI governance frameworks also present a promising pathway for aligning technological progress with ethical considerations.

Looking ahead, the future of AI governance will be shaped by the balance between innovation and regulation, the adaptability of governance frameworks to evolving AI capabilities, regulatory mechanisms that are tailored to sociocultural contexts, and the continued integration of human-centered values. As AI systems become more autonomous and embedded in critical decision-making processes, the need for robust oversight and accountability will only intensify. Ensuring responsible and human-aligned AI governance will require sustained multistakeholder engagement, proactive risk management, and a commitment to regulatory agility. By embedding ethical considerations into the very fabric of AI governance, societies can harness the benefits of AI while mitigating its potential harms, paving the way for a more equitable and just digital future.

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